Stylometric Differences and Overlap Among Human, LLM-Generated, and Machine-Translated Bangla

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Abstract

This study investigates whether stylometric features can distinguish original human-written, directly large language model (LLM)-generated, and machine-translated Bangla texts. A dataset of 300 texts was constructed as 100 topic-matched triplets comprising human-written Bangla news passages, directly LLM-generated passages, and Bangla translations produced using NLLB-200 distilled 600M. Sixteen surface-level features representing document length, sentence structure, lexical diversity, punctuation, digit use, and script composition were extracted. The analysis combined descriptive statistics, Kruskal–Wallis tests, pairwise Mann–Whitney U tests with Bonferroni correction, rank-biserial effect sizes, principal component analysis (PCA), and four supervised classifiers evaluated using five-fold topic-grouped cross-validation. Kruskal–Wallis tests indicated significant differences among the three conditions in 14 of 16 features. The first two principal components explained 53.24% of the variance, revealing broad group structure alongside visible overlap. XGBoost achieved 94.67% accuracy and macro F1 using all features; accuracy decreased to 83.67% when raw document-size and punctuation-count features were excluded. These findings suggest that the three text-production conditions exhibit distinguishable but overlapping stylometric profiles within this dataset. Potential source and formatting effects, imperfect genre control, and the use of a single LLM and machine translation system limit generalisation.

1. Introduction

Large language models (LLMs) and neural machine translation (NMT) systems can now produce fluent text in many languages, including Bangla (Bengali). Consequently, a Bangla passage encountered online may have been written originally by a person, generated directly by an LLM, or produced by translating a source text from another language. These text-production conditions have different implications for authorship analysis, content provenance, corpus construction, and language technology evaluation. Machine-translated text is particularly important to distinguish because it is expressed in Bangla but originates from a source written in another language.

Research on machine-generated text (MGT) detection has traditionally examined the distinction between human-written and machine-generated text, with multilingual benchmarks expanding the scope of this task [1, 2, 3]. However, the reliability of AI-text detection can vary with text length, domain, and generator, raising questions about its generalisability [4]. A related research tradition, translationese, investigates systematic linguistic differences between texts originally written in a target language and texts translated into it. Previous studies have shown that these differences can be captured through linguistic features and learned by classifiers [5, 6, 7]. More recent work has compared human translation, NMT output, and LLM-produced translations [8, 9, 10, 11, 12]. However, the studies identified in this review examine languages such as English, Chinese, German,

Arabic, Portuguese, and Spanish. Research on Bangla has also investigated human-versus-AI text detection, including studies of LLM-generated news and AI-generated paraphrases [13, 14, 15]. The extent to which original human-written, directly LLM-generated, and machine-translated Bangla can be distinguished within a single comparative design therefore warrants further investigation.

This study examines whether interpretable stylometric features distinguish these three text-production conditions in Bangla using 100 topic-matched triplets. Rather than developing a universal AI-text detector, we investigate measurable differences and overlap among the conditions and examine how classification performance changes when raw document-size and punctuation-count features are excluded. This reduced-feature analysis provides a preliminary assessment of whether classification remains effective when some potentially influential surface-level features are removed.

We address four research questions:

1. **RQ1:** Which stylometric features differ among human-written, directly LLM-generated, and machine-translated Bangla texts?
2. **RQ2:** To what extent do the three conditions overlap in a low-dimensional representation of their stylometric features?
3. **RQ3:** How accurately can supervised models distinguish the three text-production conditions under five-fold topic-grouped cross-validation?
4. **RQ4:** How does classification performance change when raw document-size and punctuation-count features are excluded?

This study makes four contributions: (i) a corpus of 300 Bangla texts organised into 100 topic-matched triplets across three text-production conditions; (ii) a transparent analysis of 16 interpretable stylometric features; (iii) a comparative evaluation of four supervised classifiers using topic-grouped cross-validation; and (iv) a reduced-feature analysis examining the contribution of document-size and punctuation-count features to classification performance. We present this work as an initial empirical investigation of three text-production conditions in Bangla, not as evidence that they can always be identified reliably or perfectly.

2. Related Work

2.1 Stylometry and human versus LLM-generated text

Stylometry characterises writing through measurable linguistic properties, including word and sentence length, lexical diversity, function-word frequencies, character n-grams, and punctuation. It has a long history in authorship attribution [16], including applications to Bangla [17]. More recently, stylometric features have been investigated for distinguishing human-written text from LLM-generated text. Herbold et al. compared human-written and ChatGPT-generated argumentative essays and reported systematic linguistic differences, including differences in lexical diversity and sentence complexity [18]. Kumarage et al. investigated stylometric signals for detecting AI-generated text in short Twitter timelines [19]. Zaitsu et al. examined Japanese texts written by humans and generated by seven LLMs [20], while Przystalski et al. studied the use of stylometry for distinguishing human and LLM-generated text in short samples [21]. Collectively, these studies motivate the use of interpretable linguistic features for analysing text-production patterns. However, the studies considered here do not directly address the three-way comparison of original human-written, directly LLM-generated, and machine-translated Bangla investigated in this study.

2.2 Multilingual and low-resource MGT detection

Several benchmarks have broadened machine-generated text (MGT) detection beyond English, including M4 [1], MULTITuDE [2], and SemEval-2024 Task 8 [3]. Their language coverage and task definitions provide important context for multilingual detection, although they do not establish whether the three text-production conditions examined here can be distinguished in Bangla. Other work has considered more fine-grained categories of text production. LLM-DetectAIve, for example, distinguishes human-written text from machine-generated, machine-humanised, and machine-polished English text [22], but does not explicitly include machine translation as a separate condition. Kavathekar et al. investigated the detectability of AI-generated Hindi news [23]. Chakraborty et al. discussed the limitations of AI-generated text detection as the distributions of human and machine-written text converge, highlighting the importance of factors such as sample length [4]. These findings motivate examining both classification performance and the contribution of document-length-related features, rather than interpreting detection accuracy alone as evidence of stylistic differences.

2.3 Translationese, NMT, and LLM translation

Translationese refers to systematic linguistic characteristics associated with translated text compared with text originally written in the target language. Baroni and Bernardini demonstrated that machine-learning methods could distinguish original from translated Italian [5]. Koppel and Ordan investigated both source-language-specific and more general effects of translation [6], while Volansky et al. evaluated a range of theoretically motivated features and found that their discriminative value varied [7]. Related work has examined translationese in neural machine translation (NMT) output [24] and the relationship between translationese features and translation quality [25].

More recent studies have compared human translation, NMT, and LLM-based translation. Jiang et al. classified Chinese-to-English translations produced by humans, Google Translate, and ChatGPT, reporting high classification accuracy while finding that unsupervised clustering did not clearly separate the conditions [8]. Sizov et al. compared original German text with human, NMT, and LLM translations and reported differences between machine-translated and originally authored text [9]. Fan et al. [10] and Zhang and Zhao [26] investigated interpretable features for classifying human and machine translations involving Chinese. Niu et al. further discussed machine-translated language as a distinct translational variety [27]. Work presented at the 2026 StyGenAI workshop has also examined

stylometric differences involving English–Arabic translation [12] and human-written, LLM-generated, and translated texts across English, Brazilian Portuguese, and Spanish [11].

This literature establishes that translated text can exhibit measurable linguistic patterns, but the findings cannot automatically be generalised to Bangla. It also motivates treating machine-translated Bangla as a distinct text-production condition rather than grouping it with directly LLM-generated text.

2.4 Bangla NLP and AI-generated Bangla

Bangla is spoken by a large population but remains comparatively under-resourced in several areas of natural language processing. Models such as BanglaBERT have contributed to the development of language-specific NLP resources and methods [28]. Research on detecting or analysing AI-generated Bangla has also expanded. Raya et al. introduced BanglaTextDistinguish, a dataset involving human-written and ChatGPT-paraphrased Bangla text [13]. Islam et al. evaluated transformer-based approaches on this dataset, reporting differences between zero-shot and fine-tuned performance [14]. Akther et al. investigated LLM-generated or manipulated Bangla news and analysed properties such as text length and sentence distributions alongside detection [15].

Additional resources include BanglaAiDetect, which contains human-sourced and LLM-generated texts [29], and a human-versus-AI Bangla text classification dataset [30]. These resources are relevant to Bangla AI-text research, but their task definitions and included text-production conditions differ. In particular, human-versus-AI classification and LLM-paraphrase detection do not by themselves establish how original human-written Bangla, directly LLM-generated Bangla, and machine-translated Bangla compare within one study. This distinction motivates the three-condition design adopted here.

2.5 Prior-work search and positioning

To investigate the research gap, we searched the web, ACL Anthology, arXiv, IEEE, PLOS, Frontiers, Zenodo, and Mendeley Data in October 2026. The search focused on studies combining four elements: (a) Bangla text, (b) originally human-written text, (c) directly LLM-generated text, and (d) machine-translated text. Among the studies identified in this search, we found no study that explicitly combined all four elements in a single comparative design. This finding should be understood as the result of a bounded literature search, rather than proof that no related work exists.

The closest conceptual comparison identified was Resende and Castilho [11], which examined human-written originals, ChatGPT-4o-generated originals, human translations, and ChatGPT-4o translations across three European languages. The study differs from ours in language coverage, corpus scale, and experimental design; it also investigates LLM-based translation rather than the dedicated NMT system used here. The present study therefore addresses a specific empirical question: whether interpretable stylometric features can distinguish original human-written, directly LLM-generated, and NLLB-translated Bangla within a topic-matched comparative dataset.

Table 1. Selected related studies and their relationship to the present study.

Study	Language(s)	Original human	Direct LLM	Machine translation	Method
Raya et al. 2024 [13]	Bangla	Yes	Paraphrase	No	BiLSTM-SVM
Islam et al. 2025 [14]	Bangla	Yes	Paraphrase	No	Transformers
Akther et al. 2025 [15]	Bangla	Yes	Yes (7 LLMs)	No	Analysis + detectors
Hossain 2025 [29]	Bangla	Yes	Yes (3 LLMs)	No	Dataset only

Study	Language(s)	Original human	Direct LLM	Machine translation	Method
Jiang et al. 2023 [8]	ZH→EN	No (HT)	LLM translation	Yes	Register features + ML
Sizov et al. 2024 [9]	DE/EN	Yes	LLM translation	Yes	Classifier + lexical
Fan et al. 2026 [10]	ZH→EN	No (HT)	LLM translation	Yes	Interpretable ML
Awwad et al. 2026 [12]	EN↔AR	No (HT)	LLM translation	Yes	Stylometry
Resende & Castilho 2026 [11]	EN/PT/ES	Yes	Yes	LLM only	Stylometry (12 texts)
This study	Bangla	Yes	Yes	Yes (NLLB)	Stylometry + stats + ML

HT = human translation used as the human condition (not original target-language writing).

3. Dataset

3.1 Overview

The corpus comprises 300 Bangla texts organised into 100 topic-matched triplets, with one text from each of three production conditions: original human-written Bangla, directly LLM-generated Bangla, and machine-translated Bangla. The master dataset, `final_bangla_stylometry_300.csv`, contains one row per topic with the fields `topic_id`, `topic`, `human_bangla`, `llm_bangla`, `english_passage`, and `mt_bangla`. A corresponding long-format file, `final_bangla_stylometry_long_300.csv`, contains one text per row with the fields `topic_id`, `topic`, `condition`, and `text`. The condition labels are `human`, `llm`, and `machine_translation`.

Table 2. Dataset composition.

Condition	Texts	Source / generator	Target length
Human-written Bangla	100	Bengali News Corpus (bnnews) excerpts, 2023–2026 [31]	~100–200 words
LLM-generated Bangla	100	Topic-prompted LLM [Gemini 2.5 Flash]	~100–200 words
Machine-translated Bangla	100	Infini-News English passages [32] → NLLB-200 distilled 600M [33]	Source-dependent
Total	300	100 topic-matched triplets	—

3.2 Human-written Bangla texts

The human-written condition consists of 100 passages selected from the Bengali News Corpus (bnnews), a collection of cleaned Bangla news articles from 2023–2026 that includes material from outlets such as Prothom Alo [31]. One passage of approximately 100–200 words was selected for each topic. Selection aimed to preserve the original wording while avoiding, where possible, headlines, lists, advertisements, metadata, URLs, and incomplete fragments.

Selection metadata were recorded, including topic ID, article title, selected passage, word count, category, publication date, source, URL, relevance score, and selection notes. The passages represent excerpts from news articles rather than Bangla writing in general. Furthermore, the corpus should not

be interpreted as representing 100 independent authors, because author independence was not established.

3.3 LLM-generated Bangla texts

One hundred LLM-generated Bangla texts were created using Google's Gemini 2.5 Flash model (`gemini-2.5-flash`). Each generation prompt contained the assigned topic but did not provide the corresponding human-written Bangla text or English passage. The model was instructed to produce an original, neutral, news-style Bangla paragraph of approximately 150 words, with an allowed range of 100–200 words.

Generation settings were held constant across topics: temperature 0.7, top-p 0.9, and a maximum output length of 500 tokens. An automated Bangla-script ratio check was used for quality control, with up to five generation attempts when the ratio fell below approximately 0.65. Generated texts were not manually edited.

Table 3. LLM generation and machine-translation settings.

Parameter	LLM generation	Machine translation
Model	Gemini 2.5 Flash	facebook/nllb-200-distilled-600M
Direction / input	Topic prompt → Bangla	English → Bangla (local, Kaggle T4 GPU)
Temperature / Top-p	0.7 / 0.9	— (beam search, 5 beams)
Max output tokens / length	500 tokens	300 (max sequence length)
Other	Max 5 retries; Bangla-content threshold ≈ 0.65	Batch 8; repetition penalty 1.05; length penalty 1.0
Post-editing	None	None

3.4 Machine-translated Bangla texts

The machine-translation condition comprises 100 Bangla translations of English news passages retrieved from Infini-News [32]. Infini-News provides access to a large indexed collection of processed Common Crawl news articles. The passages were selected according to their relevance to the corresponding topics and then translated from English into Bangla using `facebook/nllb-200-distilled-600M`, a distilled 600-million-parameter model [33, 34]. Translation was performed locally using a Kaggle T4 GPU, with a batch size of 8, a maximum sequence length of 300, 5-beam search, a repetition penalty of 1.05, and a length penalty of 1.0. No manual stylistic rewriting was performed.

The English passages and their word counts were retained in the dataset. Unlike the human-written and LLM-generated conditions, the machine-translated passages were not constrained to a target length of 100–200 Bangla words. Their lengths depended on the selected English sources and the translation process. Consequently, differences in document length may reflect both production condition and source-text characteristics.

The English passages were matched to the human-written passages by topic, not by original article. Some English sources also originated outside Bangladesh. Differences in source country, publication outlet, and writing conventions may therefore contribute to observed differences between conditions.

3.5 Meaning of topic matching

The corpus was organised into topic-matched triplets, with one text from each production condition for every topic. Topic matching was intended to reduce the possibility that a classifier would distinguish the conditions solely because they discussed different subjects. The texts were broadly comparable in topic and intended news or expository style, but they were not source-parallel, and genre and source characteristics were not perfectly controlled.

Topic matching therefore reduces, but does not eliminate, potential confounding between production condition and content. In particular, differences between the human-written and machine-translated passages may reflect source-language, source-country, or outlet characteristics in addition to translation effects. The results should be interpreted in light of these limitations.

3.6 Data quality, licensing, and ethics

The corpus was locked after construction. The reported quality-control procedures included the passage-selection criteria described above and the automatic Bangla-script ratio check for LLM-generated outputs. No comprehensive manual review of translation quality, factual accuracy, naturalness, or genre matching was performed. Accordingly, the dataset should be understood as a controlled comparative collection assembled using specified selection and generation procedures, rather than a manually validated parallel corpus.

The Bengali News Corpus is reported in the source documentation as being distributed under CC BY-NC-SA 4.0 [31]. Infini-News access is subject to its stated research-use conditions [32]. The applicable dataset licences, access conditions, and model terms must be checked before redistributing the complete collection, particularly because it contains excerpts from source news articles and translations of English passages.

The intended plan is to make the full dataset publicly available, subject to verification that redistribution is permitted under the relevant terms. No unrestricted redistribution permission is assumed here. If the terms do not permit release of particular source texts or derived translations, the release will need to be limited or restructured accordingly, for example by sharing permitted metadata, derived features, code, or references to the original sources instead of restricted text.

4 Methodology

4.1 Stylometric features

Sixteen interpretable surface-level features were extracted from each text. These features measure document length, sentence structure, lexical diversity, repetition, punctuation, digit use, and script composition. They do not directly measure syntax, morphology, semantic coherence, or grammaticality.

The definitions and rationales for all 16 features are provided in Table 4. Several features are related, including word count and character count, sentence count and full-stop count, and the three lexical-diversity measures. Type-token ratio is also sensitive to text length [35].

Table 4. Stylometric feature definitions.

Feature	Definition	Rationale
word_count	Number of detected words	Document length
sentence_count	Number of detected sentences	Segmentation density

Feature	Definition	Rationale
avg_sentence_length	Words / sentences	Sentence length
avg_word_length	Mean character length of non-numeric words	Word-length pattern
unique_word_count	Distinct normalised words	Vocabulary variety (length-sensitive)
type_token_ratio	Unique words / non-numeric tokens	Lexical diversity (length-sensitive)
root_ttr	Unique words / √(non-numeric tokens)	Partially length-adjusted diversity
char_count	Characters in raw text	Document size
punctuation_count	Commas + Bangla full stops + ? + !	Punctuation frequency
comma_count	Number of commas	Clause punctuation
fullstop_count	Number of Bangla full stops (দাঁড়ি, U+0964)	Sentence-final punctuation
question_count	Number of question marks	Interrogatives
exclamation_count	Number of exclamation marks	Exclamatives
digit_count	Number of digit characters	Numerical formatting
bangla_char_ratio	Bangla / (Bangla + Latin) alphabetic characters	Script composition
repeated_word_ratio	Tokens of repeated word types / non-numeric tokens	Repetition

4.2 Implementation

Feature extraction used Python regular expressions to identify Bangla-script sequences, Latin words, and numeric sequences. Words were lower-cased for lexical-diversity and repetition calculations, with numeric tokens excluded from these measures.

Sentences were segmented using Bangla full stops, question marks, and exclamation marks. The tokenizer does not perform morphological analysis, and the sentence splitter may miss other punctuation conventions.

4.3 Statistical analysis

Kruskal–Wallis tests [36] were used to examine differences among the three text-production conditions for each of the 16 stylometric features. Bonferroni correction was applied across the 16 omnibus tests to account for multiple comparisons, using an adjusted significance threshold of $p < 0.003125$. Pairwise Mann–Whitney U tests [37] were then conducted to identify differences between individual pairs of conditions, with Bonferroni correction applied across the three pairwise comparisons for each feature. Rank-biserial correlations were calculated to estimate effect sizes. Statistical significance was assessed using the relevant Bonferroni-adjusted threshold for each set of tests.

4.4 Principal component analysis

All 16 features were standardised to zero mean and unit variance before principal component analysis (PCA). The first two principal components explained 53.24% of the total variance.

PCA was used to visualise group structure and overlap. The components were not treated as direct measures of human or machine authorship.

4.5 Classification

Four classifiers were evaluated: logistic regression, random forest, a support vector machine (SVM) with an RBF kernel, and XGBoost [39, 40]. The models used the 16 stylometric features to classify each text into one of three conditions: human-written, LLM-generated, or machine-translated.

Evaluation used five-fold StratifiedGroupKFold cross-validation with shuffling and a random state of 42. Texts were grouped by `topic_id` to ensure that texts from the same topic did not appear in both training and test sets [38]. Each fold contained approximately 240 training texts and 60 test texts.

Model performance was evaluated using accuracy and macro-averaged F1 scores calculated from pooled out-of-fold predictions. With three equally represented classes, the uniform random-guessing baseline was 33.3%.

4.6 Reduced-feature ablation

To examine the contribution of selected document-size and punctuation-count features, the classifiers were retrained using eight features: average sentence length, average word length, type-token ratio, root type-token ratio, Bangla character ratio, repeated-word ratio, question count, and exclamation count.

This reduced-feature analysis is not complete length normalisation because type-token ratio and root type-token ratio remain length-sensitive, while question and exclamation counts remain raw counts. The results therefore indicate how performance changes when the specified features are excluded, rather than isolating entirely length-independent stylistic information.

5 Results

5.1 Descriptive statistics

Table 5 presents the mean values of the 16 stylometric features for the three text-production conditions. Human-written texts had the highest mean word count (151.74), followed by LLM-generated texts (134.37) and machine-translated texts (115.11). Machine-translated texts also had the longest average sentences (25.83 words), compared with 17.20 for LLM-generated texts and 14.60 for human-written texts.

LLM-generated texts had the highest type-token ratio (0.856) and root type-token ratio (9.913), alongside the lowest repeated-word ratio (0.246). They contained no digit characters on average, whereas human-written and machine-translated texts contained approximately eight digit characters per text. Machine-translated texts had a lower Bangla character ratio (0.946), indicating a greater proportion of Latin-script letters. Question and exclamation marks were uncommon across all three conditions.

These descriptive patterns indicate differences in text length, lexical diversity, repetition, and script composition. However, the group means alone do not establish whether the differences are statistically significant or whether they arise from stylistic differences rather than document length or source-related effects.

Table 5. Mean values of the 16 stylometric features across the three text-production conditions.

Feature	Human	LLM	MT
Word count	151.74	134.37	115.11
Sentence count	10.84	7.89	5.82
Avg. sentence length	14.60	17.20	25.83
Avg. word length	5.61	5.94	5.97
Unique word count	114.77	114.97	85.39
Type-token ratio	0.772	0.856	0.772
Root TTR	9.410	9.913	8.076
Character count	1012.37	942.06	805.71
Punctuation count	16.74	11.19	10.89
Comma count	5.97	3.30	5.35
Bangla full-stop count	10.71	7.89	5.34
Question-mark count	0.04	0.00	0.15
Exclamation-mark count	0.02	0.00	0.05
Digit count	8.09	0.00	8.48
Bangla character ratio	1.000	1.000	0.946
Repeated-word ratio	0.377	0.246	0.364

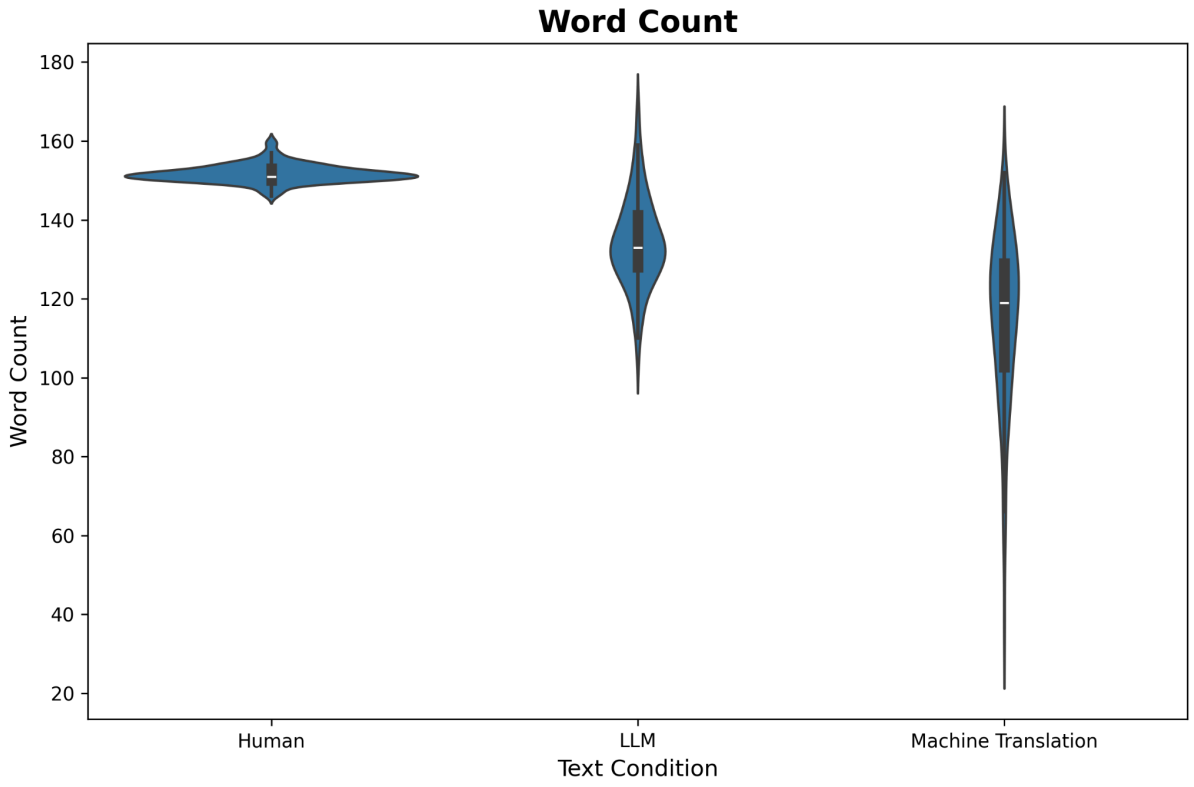


Figure 1. Distribution of word counts across human-written, LLM-generated, and machine-translated Bangla texts.

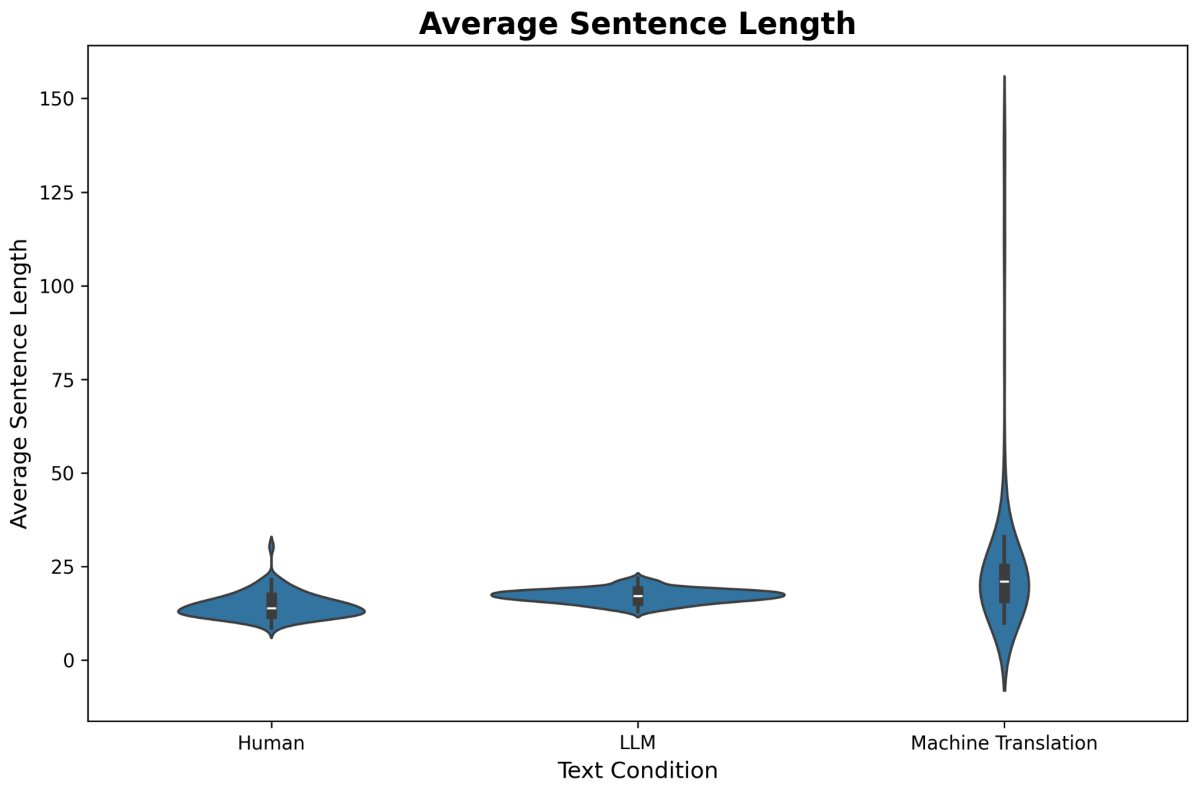


Figure 2. Distribution of average sentence length across human-written, LLM-generated, and machine-translated Bangla texts.

5.2 Inferential Comparisons

After Bonferroni correction across the 16 Kruskal–Wallis tests, 14 of the 16 stylometric features showed statistically significant differences among the three text-production conditions. Question-mark count was significant before correction (p=0.00415) but did not remain significant after correction. Exclamation-mark count was not significant before correction (p=0.130) and also remained non-significant after correction.

Pairwise Mann–Whitney U tests with Bonferroni correction were used to examine differences between each pair of conditions. Human-written and LLM-generated texts differed significantly in 12 of the 16 features, human-written and machine-translated texts differed in 10, and LLM-generated and machine-translated texts differed in 13. These results indicate that the three conditions exhibit measurable stylometric differences, while the extent of separation varies by feature and comparison.

5.3 Principal component analysis

The first two principal components explained 34.35% and 18.89% of the total variance, respectively, accounting for 53.24% combined. In the two-dimensional projection, human-written texts tended to occupy the upper-right region, LLM-generated texts tended toward the lower-right region, and machine-translated texts were more broadly distributed toward the left.

The groups nevertheless overlapped, indicating that the first two principal components did not completely separate the three conditions. This visualization provides a descriptive view of multivariate differences rather than a standalone test of classification performance.

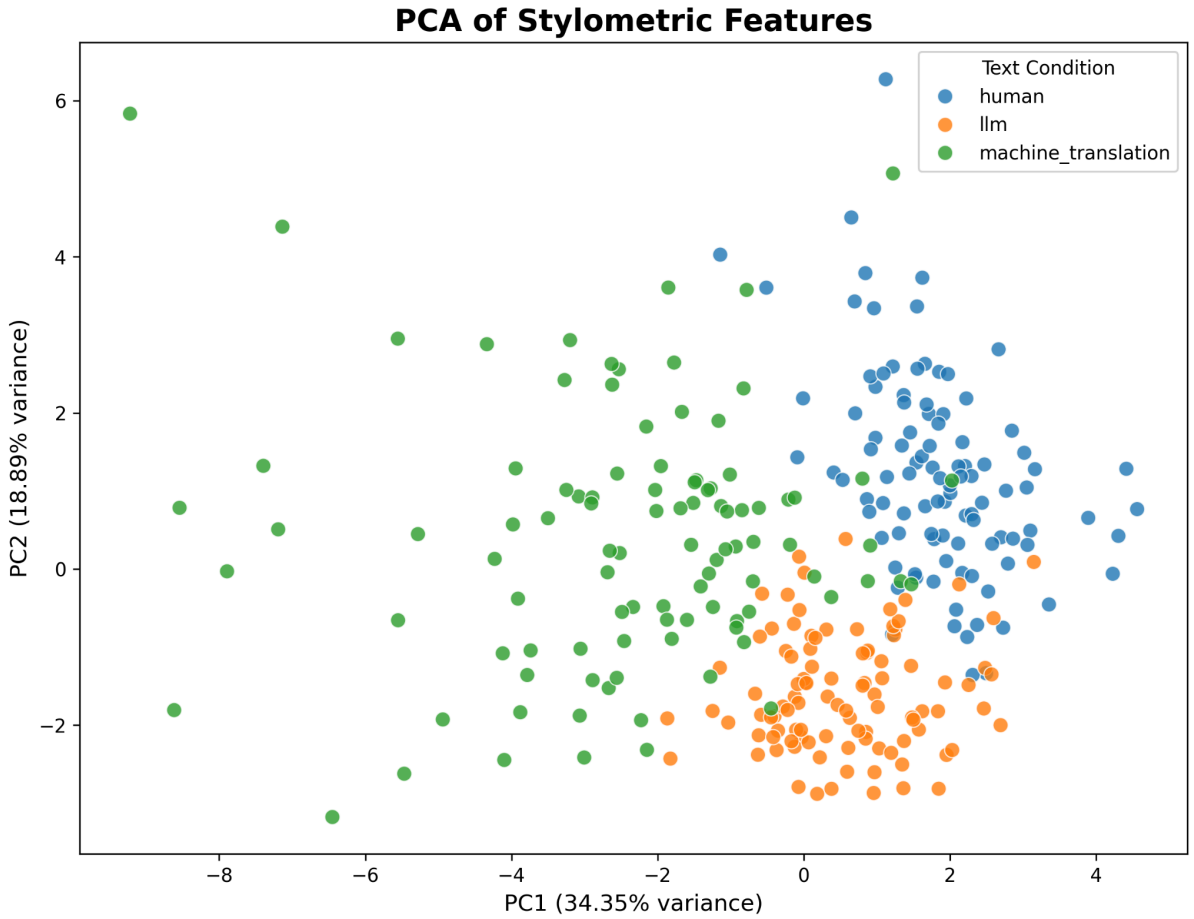


Figure 3. Principal component analysis (PCA) projection of the 300 Bangla texts using the standardised stylometric features, illustrating patterns of separation and overlap among the three text-production conditions.

5.4 Full-feature classification

Among the four classifiers evaluated using the full set of 16 stylometric features, XGBoost achieved the highest performance, with an accuracy of 94.67% and a macro F1-score of 94.67%. Random Forest followed with an accuracy of 94.33% and a macro F1-score of 94.31%, while Logistic Regression and the RBF-kernel Support Vector Machine achieved accuracies of 93.67% and 93.33%, respectively. All four classifiers performed substantially above the 33.3% uniform random-guessing reference for the three balanced classes.

XGBoost feature importance indicated that unique word count, word count, and digit count were the three most influential features, followed by full-stop count, type–token ratio, and Bangla-character ratio. These results suggest that lexical diversity, document length, and numerical content contributed substantially to the model's predictions. However, feature importance reflects the model's use of these features and does not establish that they independently cause the observed differences. The full feature-importance values are provided in Appendix B.

Table 6. Five-fold topic-grouped cross-validation results with all 16 features.

Model	Accuracy (%)	Macro F1 (%)
Logistic Regression	93.67	93.68
Random Forest	94.33	94.31
SVM (RBF)	93.33	93.31
XGBoost	94.67	94.67

Table 7. XGBoost confusion matrix, full feature set (pooled out-of-fold predictions).

Actual \ Predicted	Human	LLM	MT
Human	97	2	1
LLM	4	94	2
MT	3	4	93

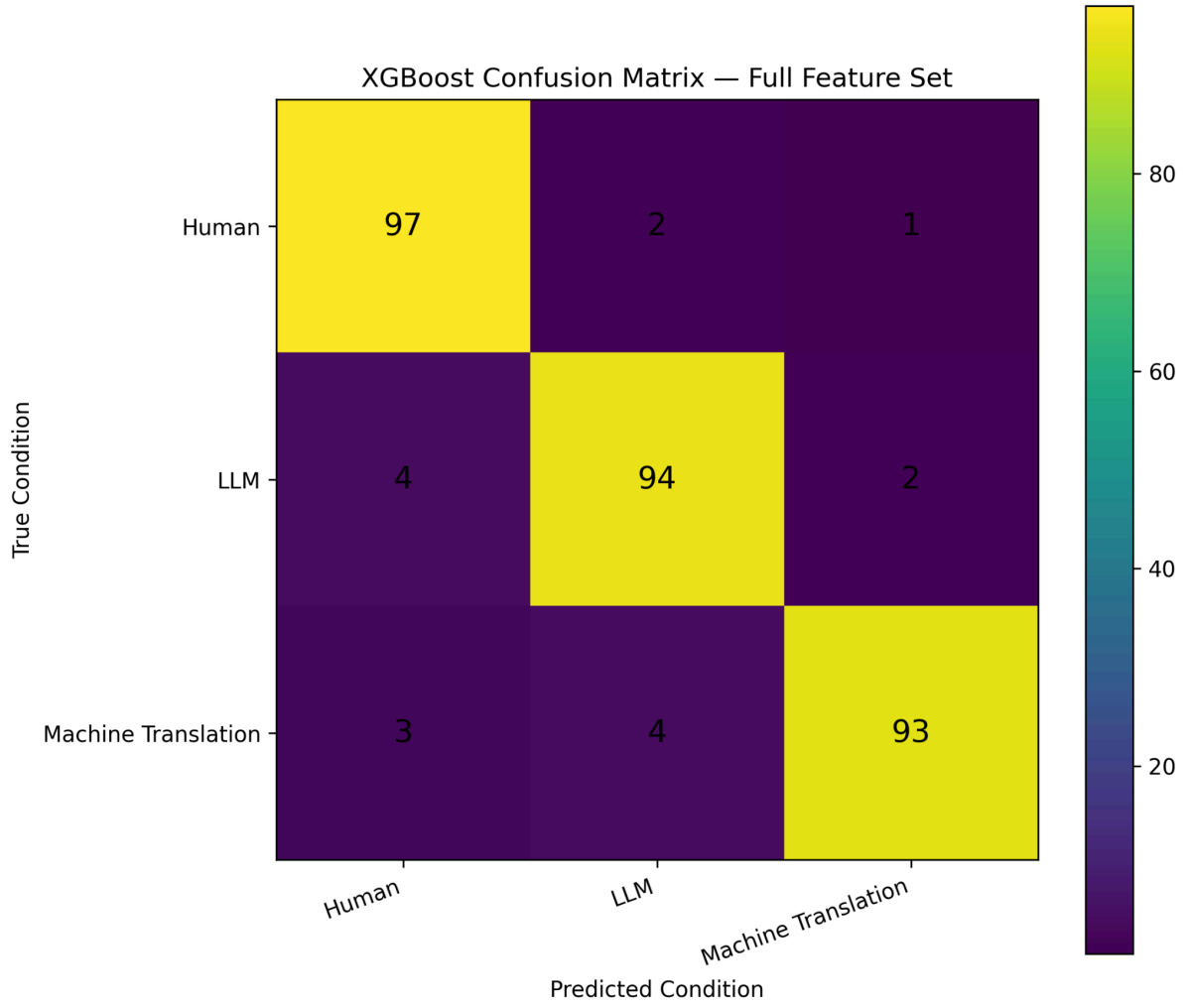


Figure 4. Confusion matrix for the full-feature XGBoost classifier, showing the classification of human-written, LLM-generated, and machine-translated Bangla texts.

Table 8. Per-class precision, recall, and F1 for the full-feature XGBoost model.

Class	Precision	Recall	F1
Human	0.933	0.970	0.951
LLM	0.940	0.940	0.940
MT	0.969	0.930	0.949

The three most important XGBoost features were unique word count (0.184), word count (0.170), and digit count (0.154), which together account for about 50.8% of total importance (Figure 6; full values in Appendix C). These are largely document-size and formatting features, indicating that the full-feature result should not be read as evidence that the model learned only linguistic style. Tree-based importance reflects model reliance, not causation, and correlated features can share or redistribute importance.

5.5 Reduced-feature ablation

Removing raw document-size and punctuation-count features lowered accuracy for every model, by 7.0 to 11.0 percentage points. Logistic regression and the SVM were most robust (86.33%), while XGBoost showed the largest drop (94.67% → 83.67%). All reduced models remained far above chance.

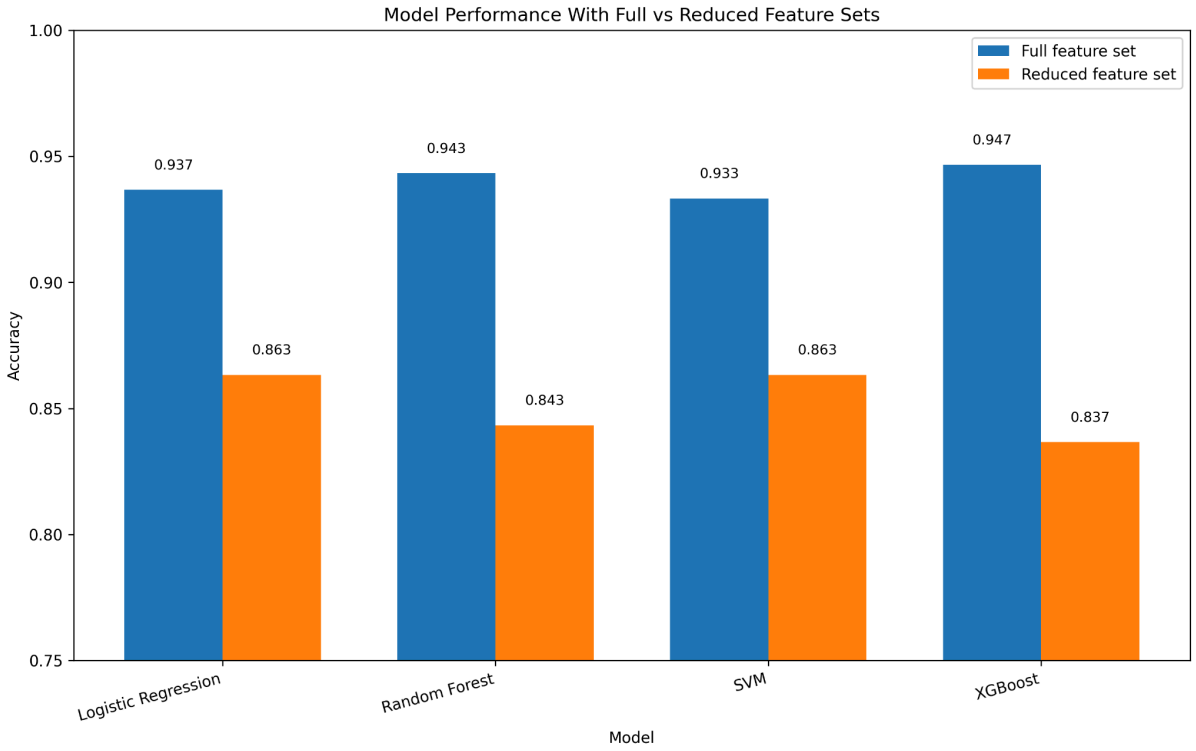


Figure 5. Classification performance of the four machine-learning models using the full and reduced feature sets, measured by accuracy and macro-averaged F1 score.

Table 9. Full versus reduced feature-set performance.

Model	Full acc. (%)	Reduced acc. (%)	Change (pp)	Reduced macro F1 (%)
Logistic Regression	93.67	86.33	−7.34	86.37

Model	Full acc. (%)	Reduced acc. (%)	Change (pp)	Reduced macro F1 (%)
Random Forest	94.33	84.33	−10.00	84.31
SVM (RBF)	93.33	86.33	−7.00	86.34
XGBoost	94.67	83.67	−11.00	83.72

The reduced XGBoost confusion matrix shows more overlap than the full model, especially between human and LLM texts: 16 human texts were predicted as LLM and 12 LLM texts as human. MT remained the most separable class (87/100 correct).

Table 10. XGBoost confusion matrix, reduced feature set.

Actual \ Predicted	Human	LLM	MT
Human	79	16	5
LLM	12	85	3
MT	8	5	87

In the reduced model, root TTR (0.222), question count (0.166), and type-token ratio (0.148) received the highest importance, followed by Bangla character ratio, average sentence length, and repeated-word ratio. The high importance of question count should be read cautiously because question marks were rare in all groups.

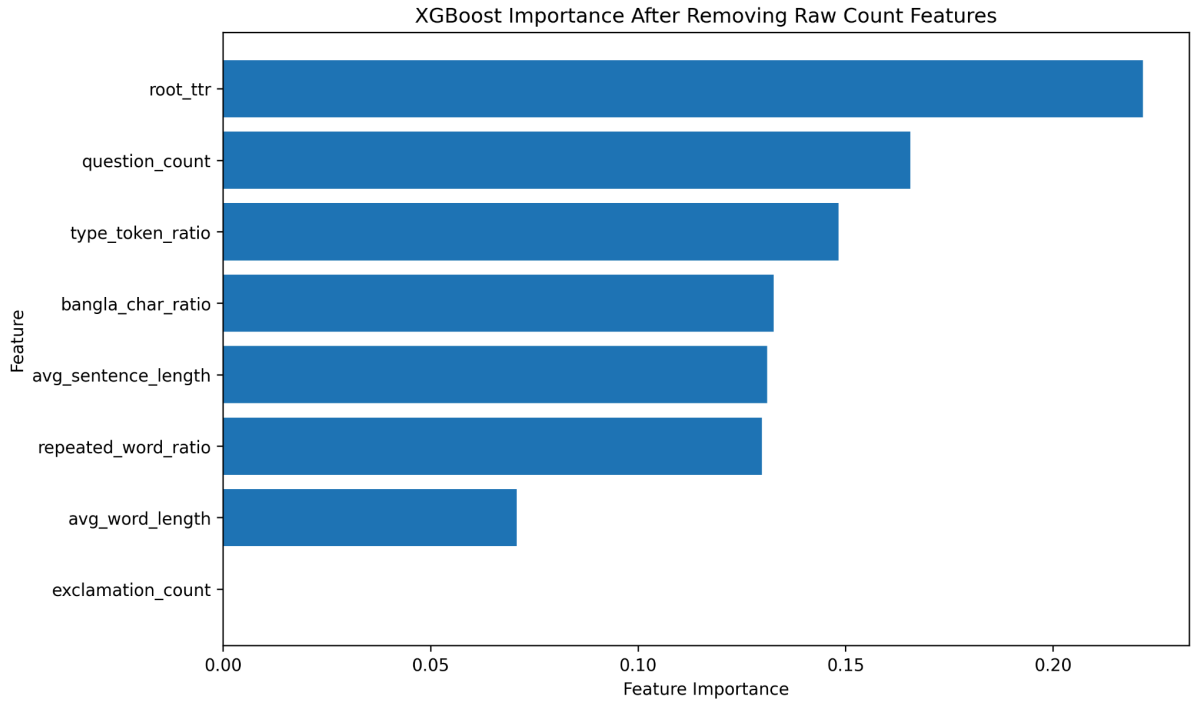


Figure 6. Feature importance scores for the reduced-feature XGBoost classifier, highlighting the stylometric features that contributed most to its predictions.

6 Discussion

6.1 Distinct but Overlapping Profiles

The descriptive statistics, PCA, and classification results indicate measurable differences among the three text-production conditions (RQ1–RQ2). Human news excerpts were longer on average and contained more punctuation, while directly generated LLM passages showed higher lexical diversity and lower repetition. Machine-translated passages had fewer, longer sentences on average and a lower Bangla character ratio, indicating a greater proportion of Latin-script letters.

The higher lexical diversity observed in LLM-generated texts is consistent with findings from English essays [18] and English, Portuguese, and Spanish stories [11]. However, direct comparisons are limited by differences in genre, text length, and lexical-diversity measures. The longer sentences in the machine-translated condition are compatible with research suggesting that translation can leave systematic traces in target-language text [7, 9, 27]. In this dataset, however, sentence length may also reflect the structure of the English source passages and the translation process.

6.2 Interpreting High Classification Accuracy Cautiously

The full-feature models achieved approximately 93–95% accuracy (RQ3). However, unique word count, word count, and digit count were the three most important features in the full-feature XGBoost model, indicating that document size and formatting contributed substantially to classification. The absence of digit characters in the LLM-generated texts is particularly notable, but it may reflect the generation prompt and model behaviour in this dataset rather than a general property of LLM-generated Bangla.

The reduced-feature analysis provides a useful comparison (RQ4). Accuracy declined by 7–11 percentage points, although the models still achieved approximately 84–86% accuracy. This suggests that features beyond raw document size and punctuation counts also contributed to classification. Nevertheless, the reduced set retained measures such as type-token ratio and root type-token ratio that remain sensitive to text length. The results therefore do not establish that classification depends solely on stylistic characteristics independent of document length or dataset composition, a concern also discussed in previous detection research [4].

6.3 Translation as a Separate Production Condition

In the reduced-feature analysis, machine-translated texts were classified correctly more often than the other classes by the XGBoost model, with 87 of 100 texts correctly identified. This suggests that the machine-translated condition had a distinguishable feature profile in the present dataset. It supports examining machine-translated text separately from directly LLM-generated text rather than assuming that both necessarily share the same measurable characteristics.

Jiang et al. [8] reported differences in the detectability of human translations, neural machine translation, and ChatGPT-generated texts. However, their findings and the present results arise from different experimental settings and should not be treated as directly equivalent. Furthermore, the English source passages used for machine translation were not source-matched to the human-written Bangla excerpts. Consequently, some observed differences may reflect source-text characteristics or corpus composition in addition to translation-related effects.

6.4 Overlap as a Finding

The PCA projections and classification confusion matrices show that the three conditions overlap in their measured features. Although group-level differences were evident, no individual feature or

combination of features should be assumed to provide an exclusive signature of a text's production condition.

These findings caution against treating stylometric classification as definitive evidence of authorship or production history. The reported performance applies to the dataset and evaluation design used in this study; broader claims would require validation on texts from additional sources, topics, models, and translation systems. In practical applications, stylometric evidence should therefore be interpreted alongside contextual information and uncertainty rather than used as a stand-alone forensic detector.

7 Limitations and Future Work

This study has several limitations. First, the dataset contains 300 texts across 100 topics, providing an initial comparison rather than broad coverage of Bangla writing across domains, regions, and genres. Second, topic matching does not control for source text, news outlet, country, genre, or formatting, so these factors may contribute to the observed differences.

Third, the full-feature classifier relied heavily on word count, unique word count, and digit count. These features may reflect document length and the generation or collection procedures rather than stylistic differences alone. Fourth, the study used one LLM configuration and one machine-translation model, limiting the extent to which the results can be generalised to other systems and settings.

Fifth, the 16 features primarily capture surface-level characteristics and do not directly represent syntax, morphology, discourse structure, or semantics. Finally, although topic-grouped cross-validation reduces the risk of topic leakage, it does not establish generalisation to unseen sources, generators, translation systems, or real-world texts with unknown production conditions. The exploratory Kruskal–Wallis and Mann–Whitney analyses also do not explicitly model the topic-level matching.

Future work should expand the dataset across more topics, sources, genres, and regions, and include multiple LLMs and machine-translation systems, including LLM-based translation. Matching text lengths and source characteristics would help separate production-condition effects from corpus differences. Further work could also investigate length-normalised feature rates, length-robust lexical-diversity measures such as MTLD [35], morphological and syntactic features, and character- and word-level n-grams. Comparisons with Bangla language models such as BanglaBERT [28], evaluation on external datasets such as BanglaAiDetect [29], and reporting fold-level performance variability would help establish the robustness and generalisability of the findings.

8 Conclusion

This study examined 300 topic-matched Bangla texts across three production conditions: human-written, directly LLM-generated, and machine-translated. Using 16 interpretable stylometric features, the analysis identified differences in document length, sentence structure, lexical diversity, repetition, punctuation, digit use, and script composition, alongside visible overlap in the PCA projection.

Under five-fold topic-grouped cross-validation, the best full-feature classifier achieved 94.67% accuracy and macro F1. Accuracy ranged from 83.67% to 86.33% across the reduced-feature models

after raw document-size and punctuation-count features were removed. These results indicate that measurable features can distinguish the three conditions within this dataset, but the contribution of document length, formatting, and potential source-related confounds limits stronger conclusions.

The findings provide preliminary evidence that human-written, directly LLM-generated, and machine-translated Bangla can exhibit distinguishable yet overlapping stylometric profiles. They do not establish a universal or fully reliable method for identifying how an arbitrary Bangla text was produced. Broader evaluation across sources, models, and genres is needed before drawing conclusions about real-world use.

Ethics Statement

The study analysed news text and machine-generated text without recruiting human participants. The research is intended to investigate measurable differences among text-production conditions and to support research on Bangla language technology, not to justify penalising individuals based on automated predictions. Stylometric classification should not be treated as conclusive evidence of authorship or used as the sole basis for consequential decisions.

Data and Code Availability

The feature-extraction and analysis code, along with the derived feature dataset, will be made available through a public repository. Redistribution of the underlying source texts will depend on the applicable dataset licences and access terms.

Acknowledgements

No external funding or individual assistance is acknowledged in this study.

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Appendix A LLM Generation Prompt

আপনি একজন পেশাদার বাংলা সংবাদ লেখক।

নিচের বিষয়টি নিয়ে একটি সম্পূর্ণ নতুন, মৌলিক বাংলা সংবাদধর্মী অনুচ্ছেদ লিখুন।

বিষয়:
{topic}

নির্দেশনা:

1. শুধুমাত্র বাংলায় লিখবেন।
2. প্রায় ১৫০টি শব্দ লিখবেন।
3. মোট শব্দসংখ্যা ১০০ থেকে ২০০-এর মধ্যে রাখবেন।
4. স্বাভাবিক, পরিষ্কার এবং প্রমিত বাংলা ব্যবহার করবেন।
5. সংবাদ প্রতিবেদনধর্মী নিরপেক্ষ ভাষা ব্যবহার করবেন।
6. বিষয়টির ওপর তথ্যভিত্তিক সাধারণ সংবাদধর্মী লেখা তৈরি করবেন।
7. কোনো শিরোনাম লিখবেন না।
8. কোনো bullet point বা numbered list ব্যবহার করবেন না।
9. কোনো markdown ব্যবহার করবেন না।
10. কোনো URL, citation, source attribution বা reference লিখবেন না।
11. এই লেখাটি সম্পূর্ণ নতুনভাবে তৈরি করবেন। কোনো বিদ্যমান লেখা অনুকরণ বা পুনর্লিখন করবেন না।
12. শুধু চাওয়া বাংলা অনুচ্ছেদটি লিখবেন। অন্য কোনো ব্যাখ্যা দেবেন না।

বিষয়টি অবশ্যই লেখাটির মূল বিষয় হতে হবে.

Appendix B Feature Importances

Table 11. XGBoost feature importances: full (16 features) and reduced (8 features) models.

Feature	Full model	Reduced model
Unique word count	0.1837	—
Word count	0.1701	—
Digit count	0.1537	—
Bangla full-stop count	0.0697	—
Type-token ratio	0.0648	0.1484
Bangla character ratio	0.0565	0.1326
Root TTR	0.0556	0.2217
Sentence count	0.0549	—
Repeated-word ratio	0.0477	0.1298
Punctuation count	0.0404	—
Avg. sentence length	0.0239	0.1311
Comma count	0.0237	—
Question count	0.0226	0.1656
Character count	0.0181	—
Avg. word length	0.0146	0.0708

Feature	Full model	Reduced model
Exclamation count	0.0000	0.0000